



Dual numbers & automatic differentiation

Martin Benning

ENGF0034 05.12.2025



Introduction

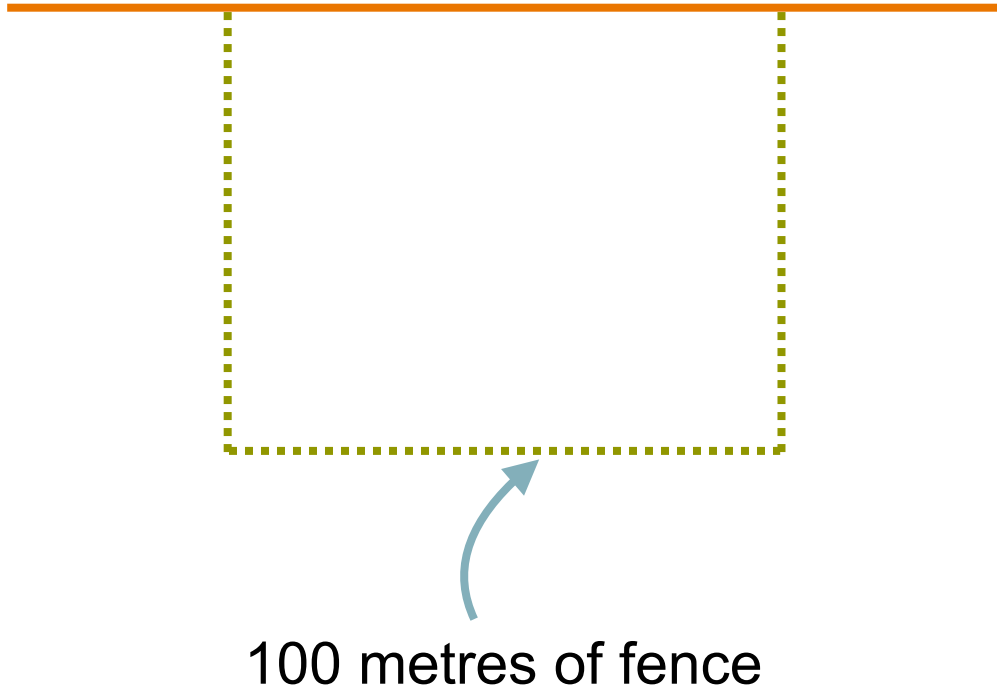
It's long ago, but one thing I particularly remember from my time as a school student:



“You have 100 metres of fence, and you want to fence off a rectangular area for your pet rabbit. One side of the enclosure is against an existing wall, so you only need fencing for the other three sides. How can you maximise the area of the enclosure?”

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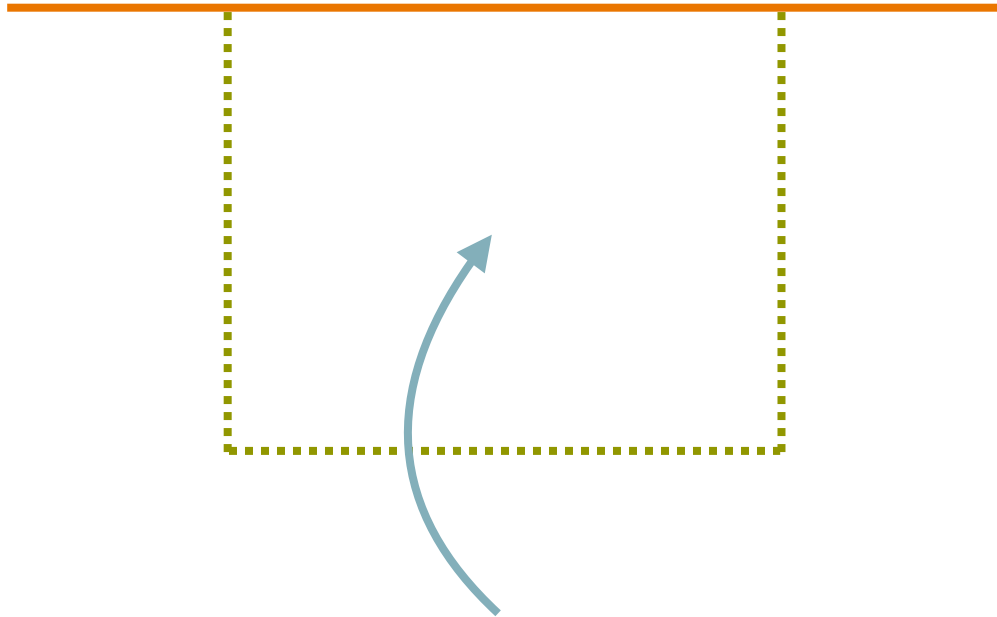
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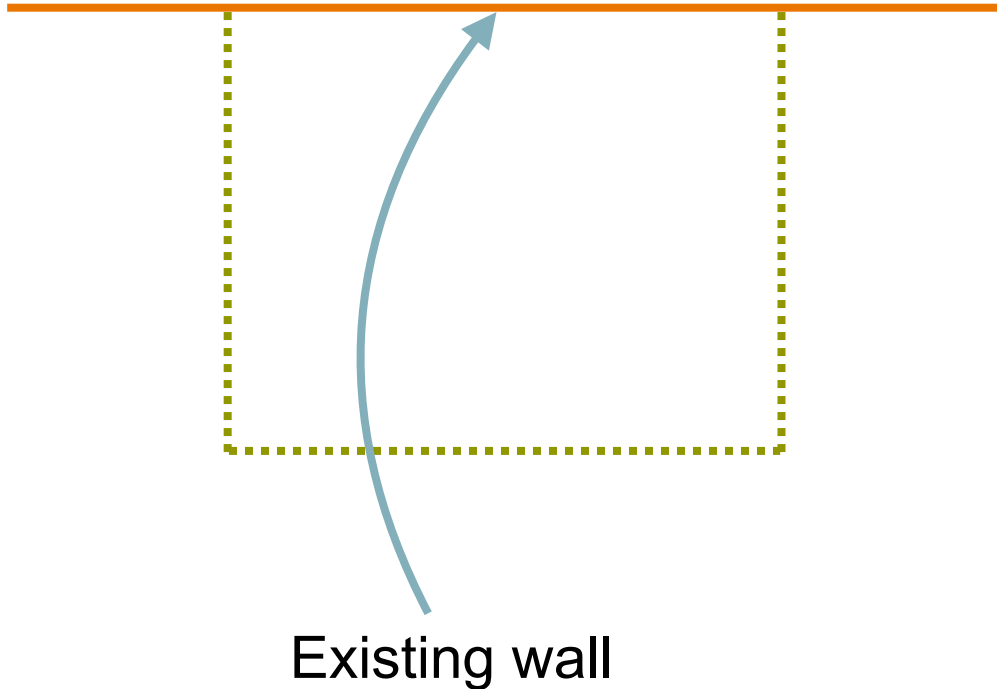


Pet rabbit enclosure

“You have 100 metres of fence, and you want to fence off a **rectangular area** for your pet rabbit. One side of the enclosure is against an existing wall, so you only need fencing for the other three sides. How can you maximise the area of the enclosure?”

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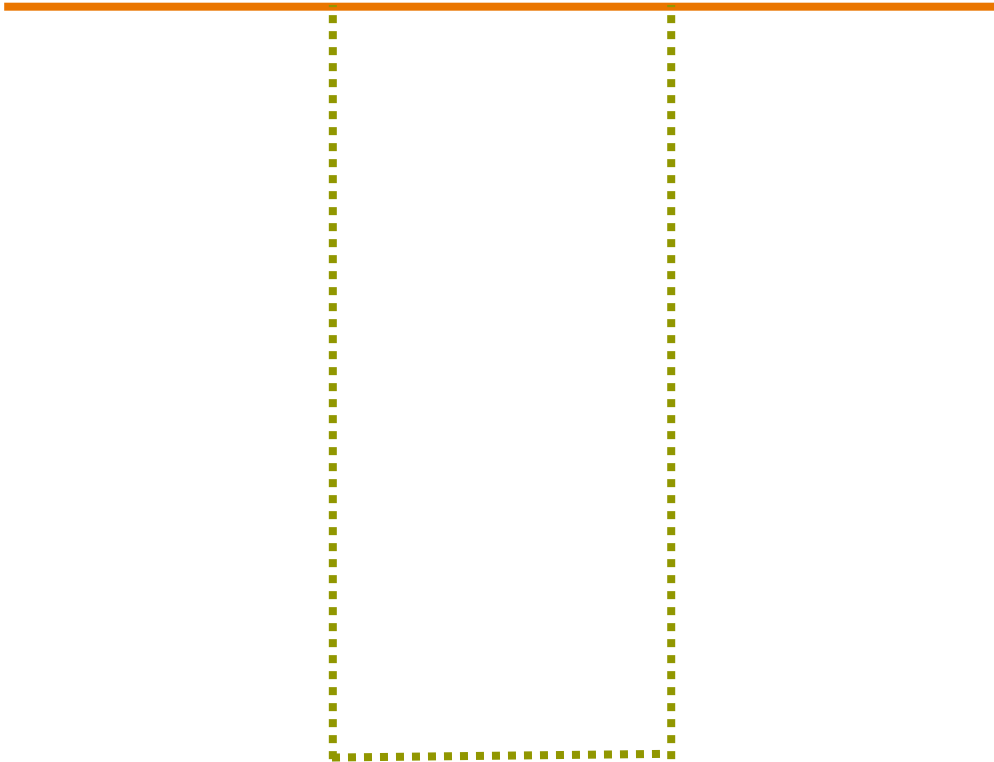
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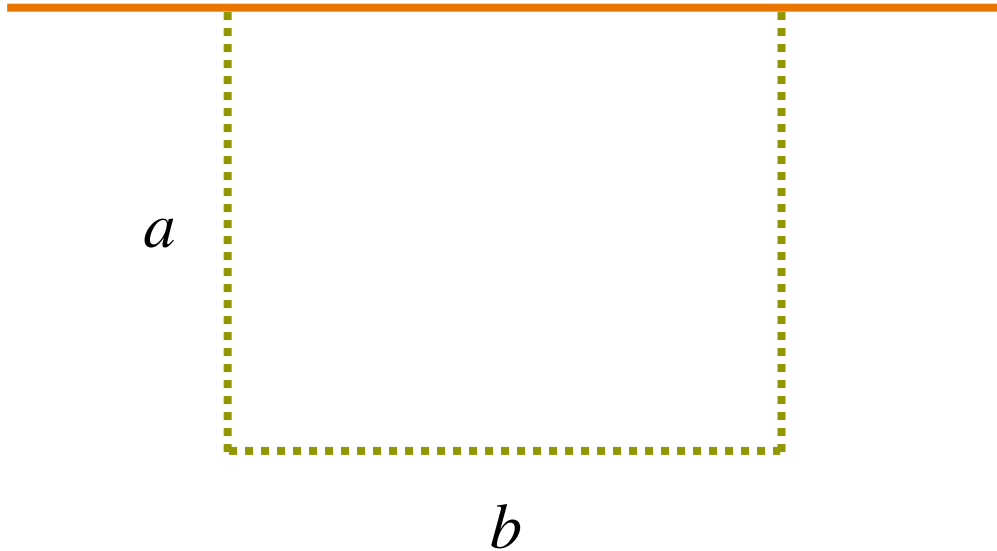
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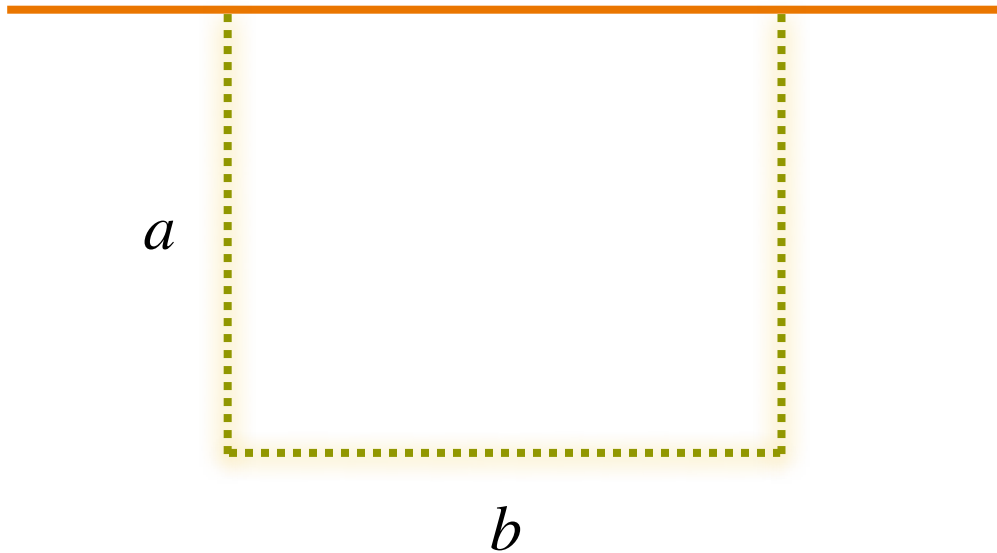
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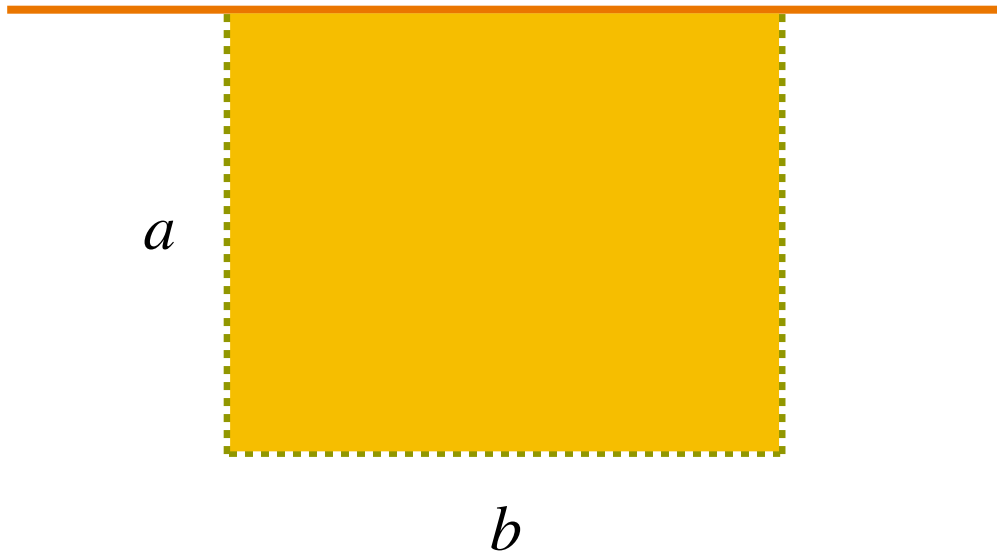


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Length of the fence: $2a + b = 100$

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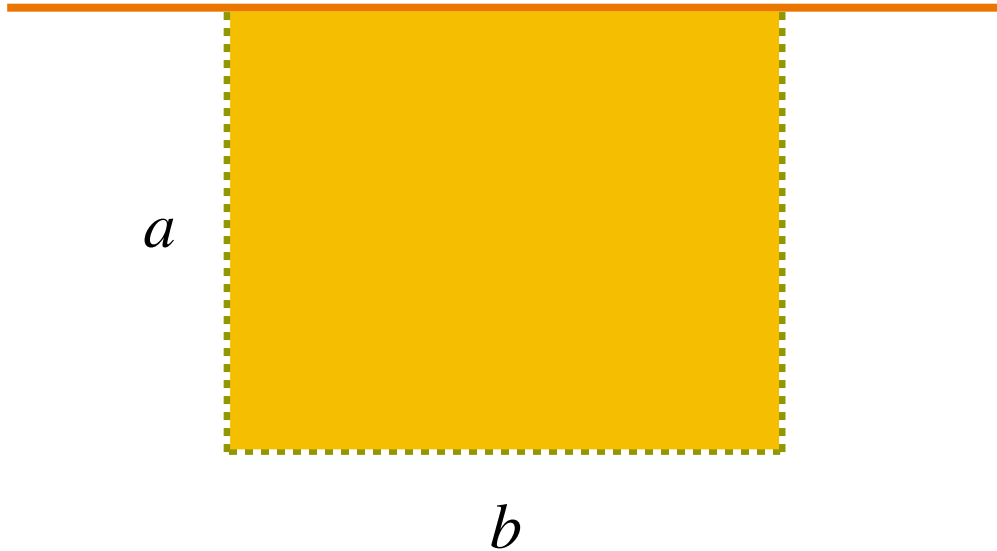
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Area of the enclosure: ab

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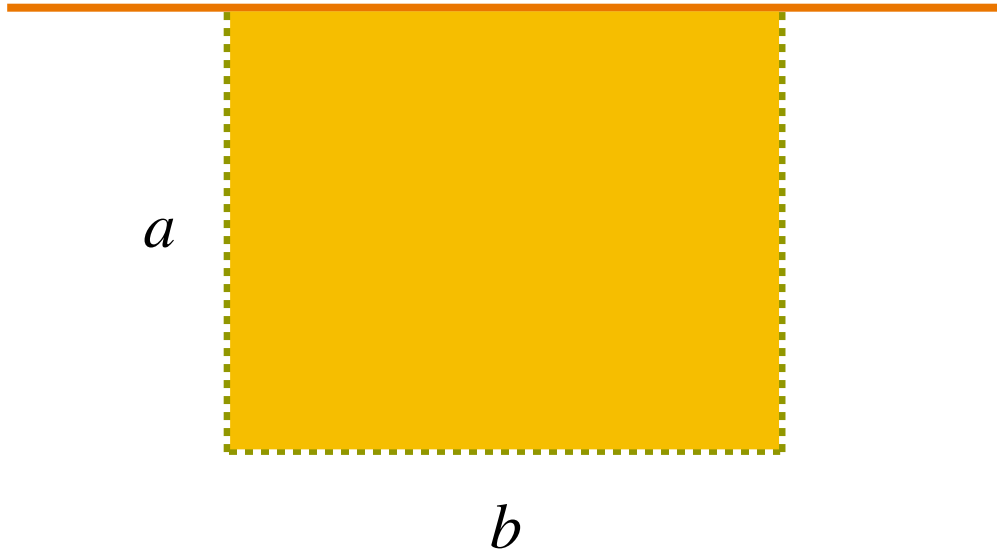
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Length of the fence: $b = 100 - 2a$

Area of the enclosure: $a b$

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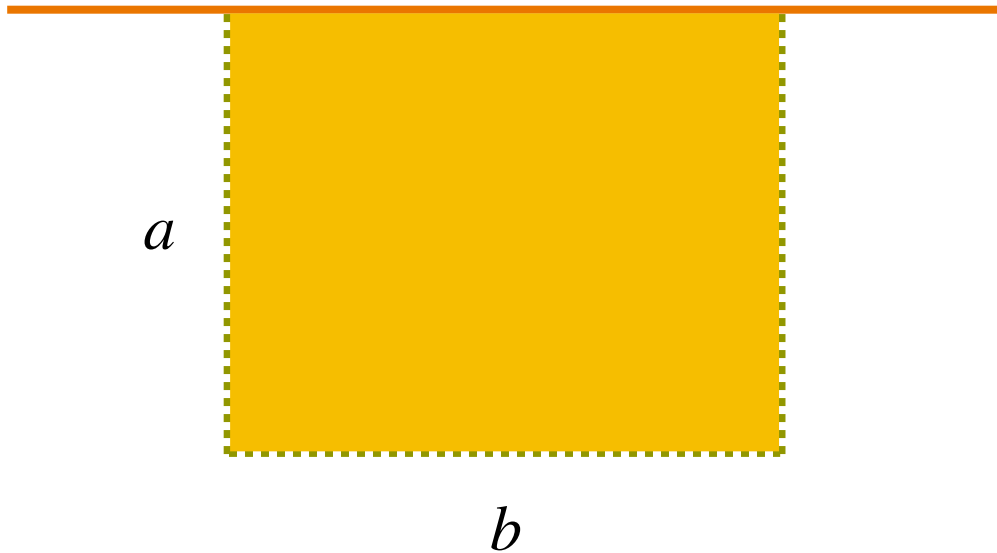


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$$\text{Area of the enclosure: } a(100 - 2a) = 100a - 2a^2 = f(a)$$

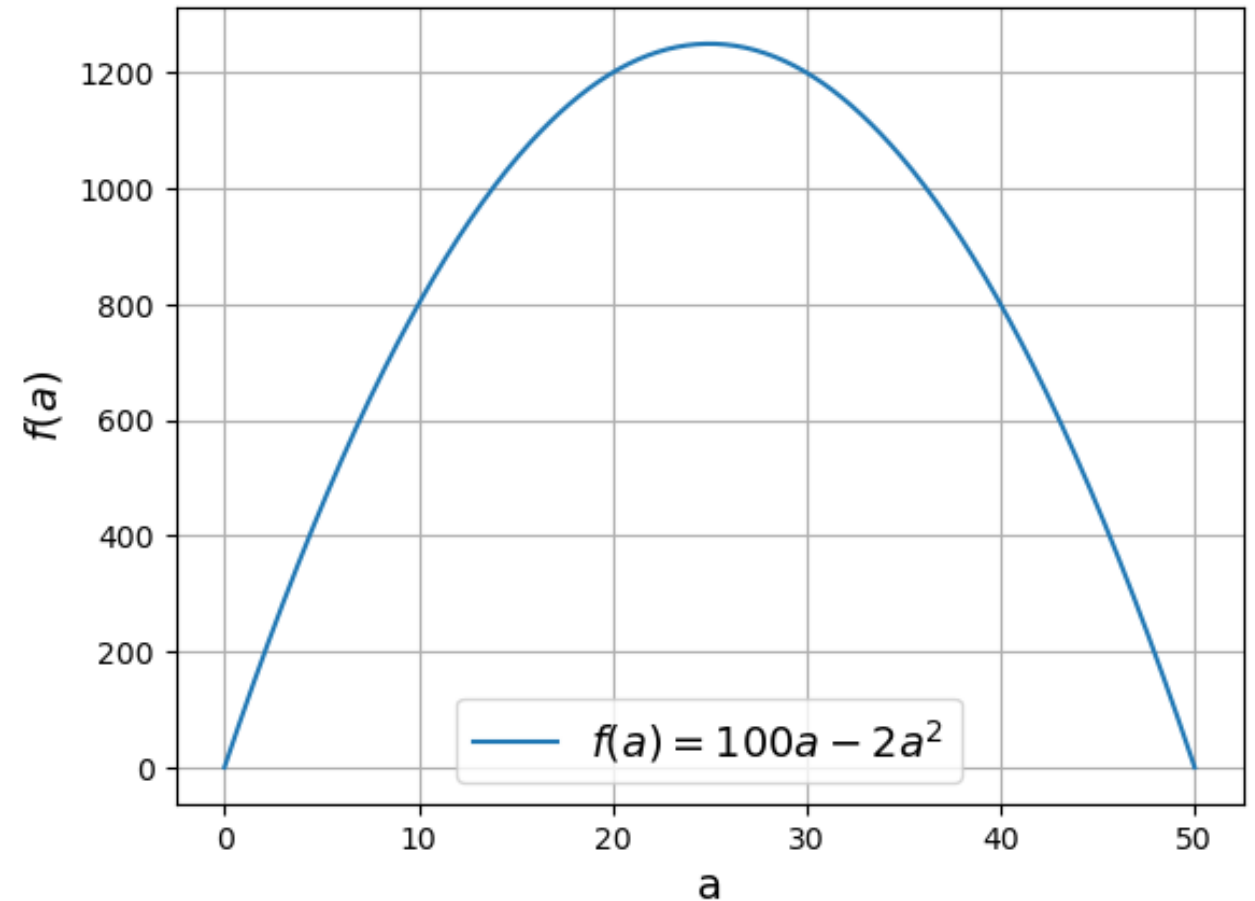
Introduction

How can you maximise the area of the enclosure?



$$f(a) = 100a - 2a^2$$

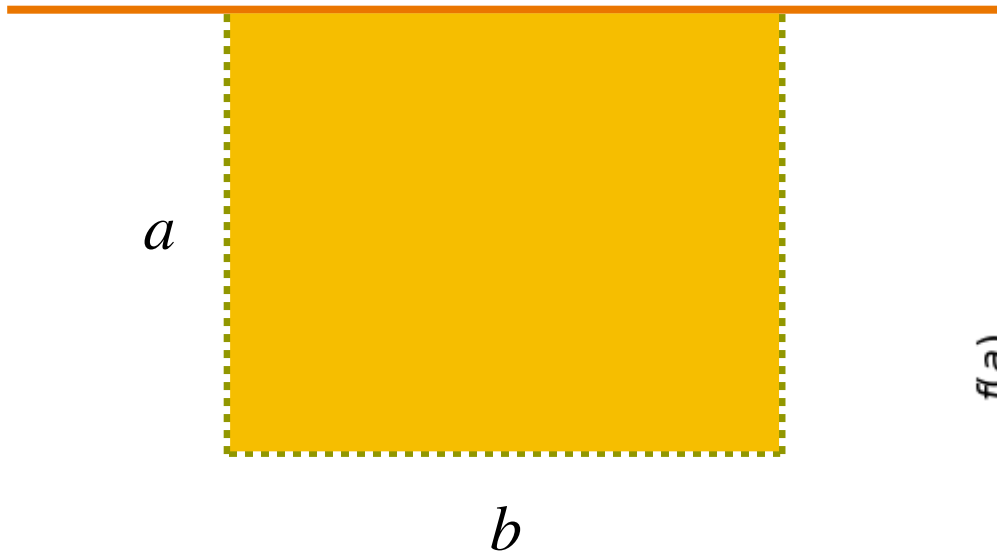
Graph of the function $f(a) = 100a - 2a^2$



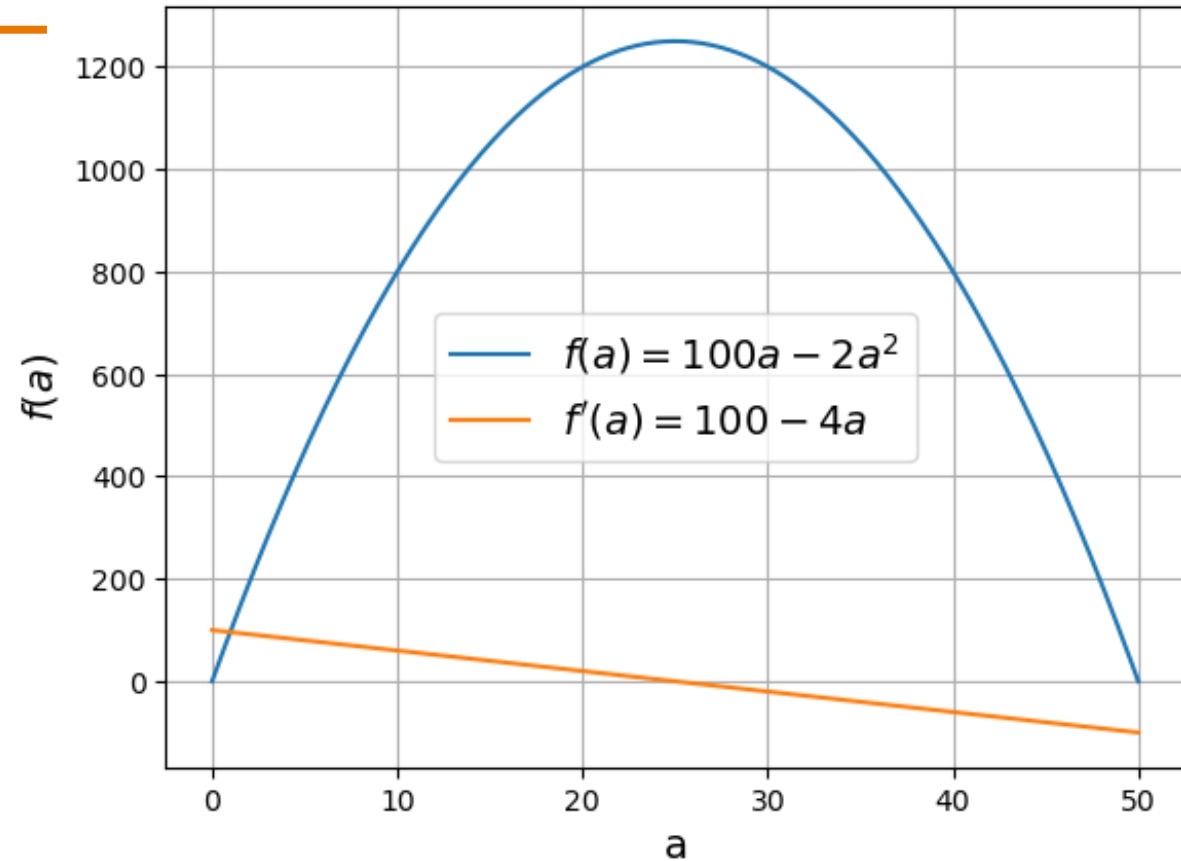
Introduction

How can you maximise the area of the enclosure?

Graph of the function $f(a) = 100a - 2a^2$ and its derivative $f'(a) = 100 - 4a$



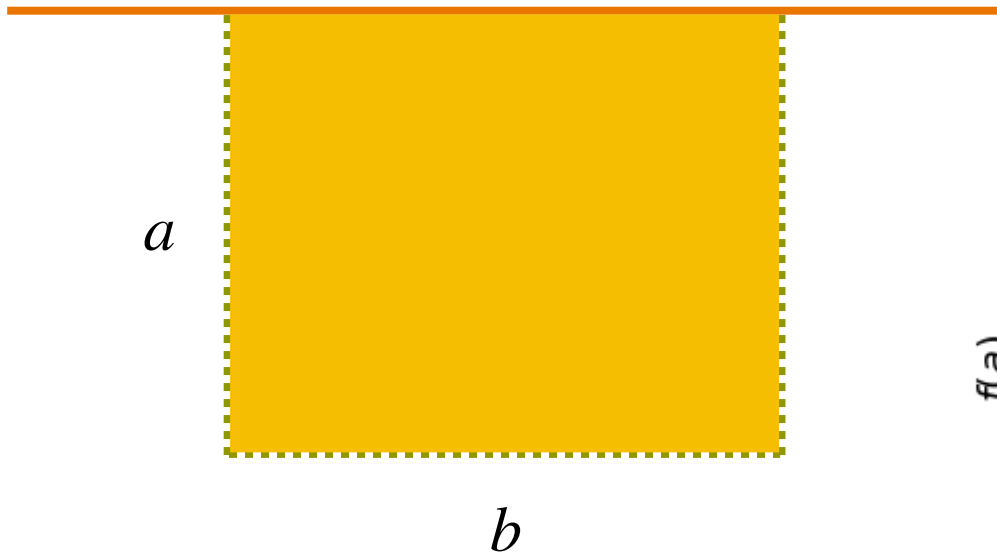
$$f(a) = 100a - 2a^2$$
$$f'(a) = 100 - 4a$$



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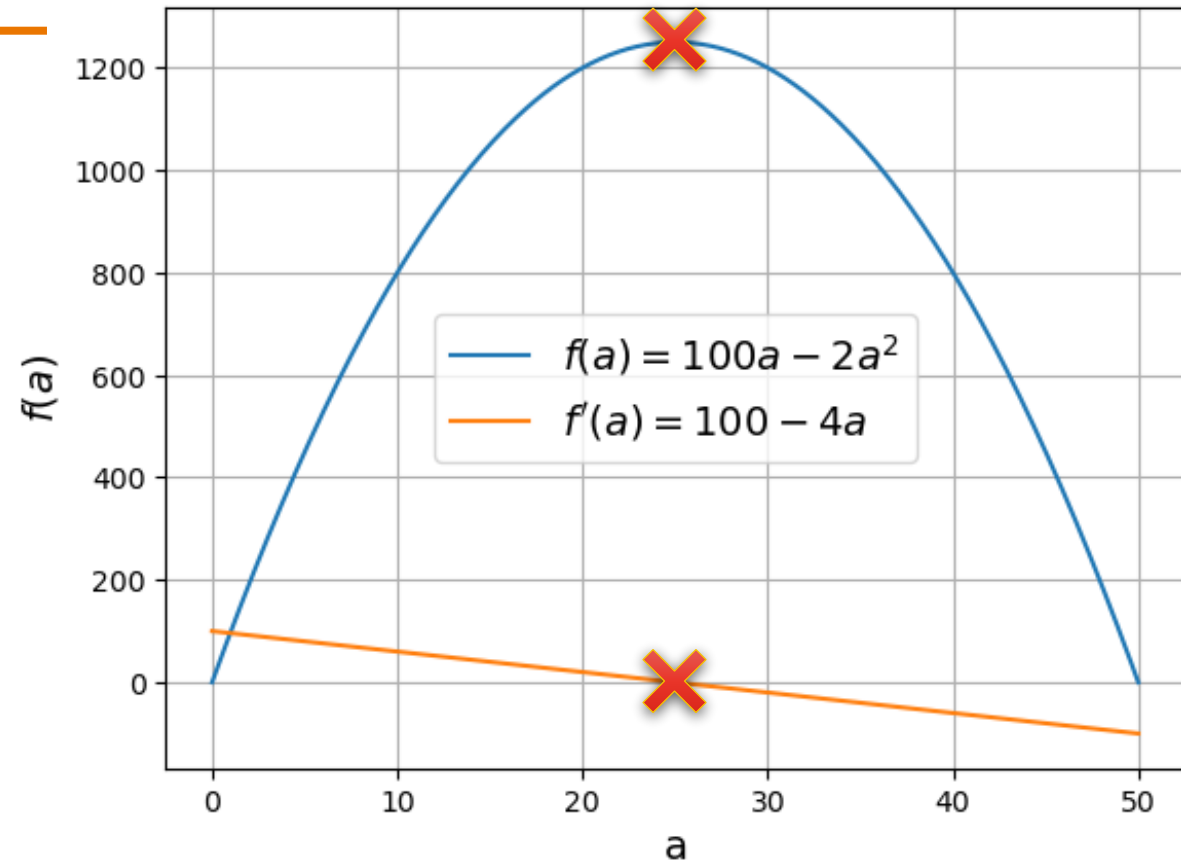
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Graph of the function $f(a) = 100a - 2a^2$ and its derivative $f'(a) = 100 - 4a$



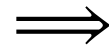
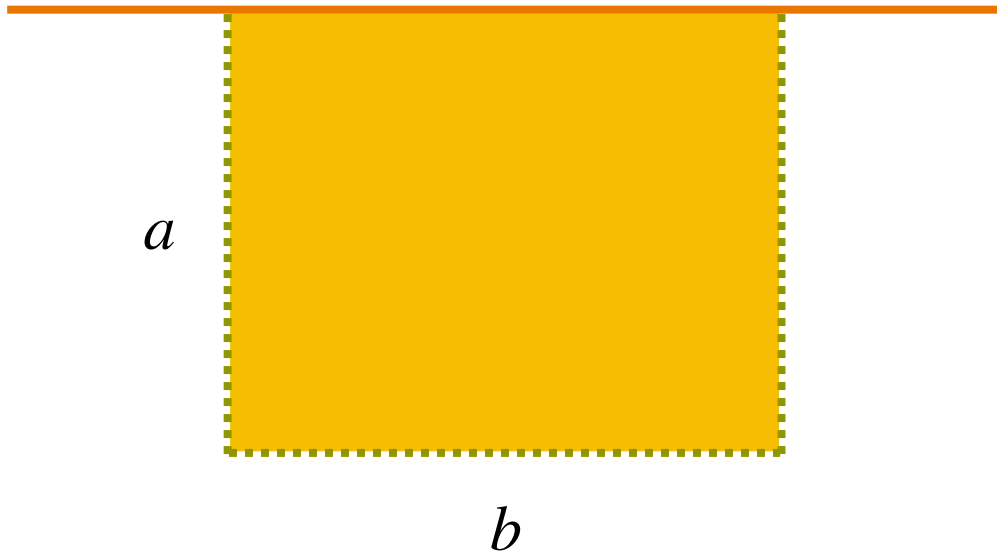
$$f(a) = 100a - 2a^2$$

$$f'(25) = 100 - 4(25) = 0$$



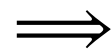
Introduction

How can you maximise the area of the enclosure?



$$\begin{aligned}a &= 25 \text{ m} \\ b &= 50 \text{ m}\end{aligned}$$

$$\begin{aligned}f(a) &= 100a - 2a^2 \\ f'(25) &= 100 - 4(25) \stackrel{!}{=} 0\end{aligned}$$



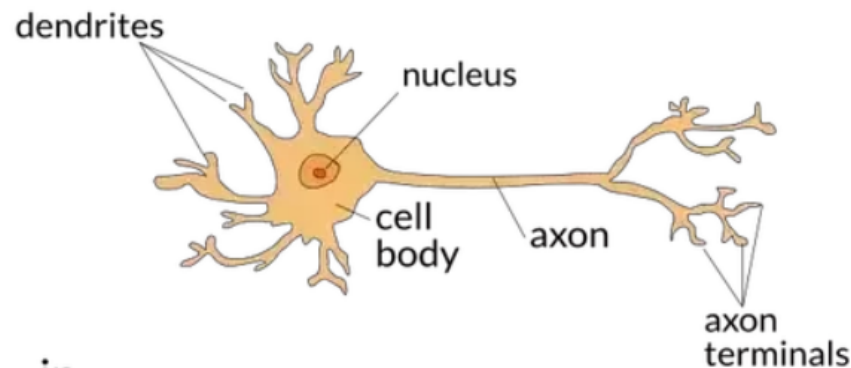
$$\text{Maximal area of enclosure} = 1250 \text{ m}^2$$

Introduction

This is maybe a very simple optimisation problem, but optimisation problems are everywhere

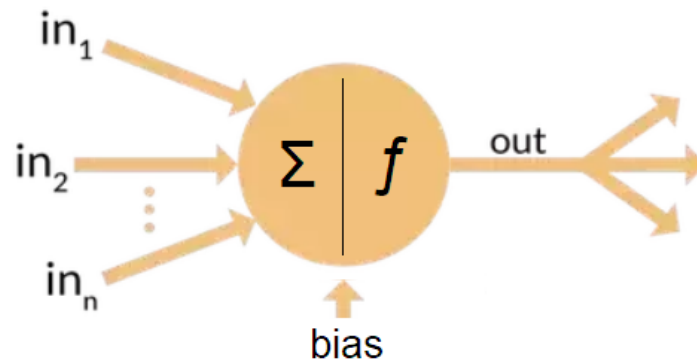
Example: training neural networks

Biological neuron



McCulloch, Pitts,
Rosenblatt, 1950s - 60s

Artificial neuron



Simplistic artificial neuron:

Neuron output $f(w, b) = \varphi(wx + b)$

Weight w Neuron input x Threshold/bias b

Activation function φ

e.g. $\varphi(z) = \tanh(z)$

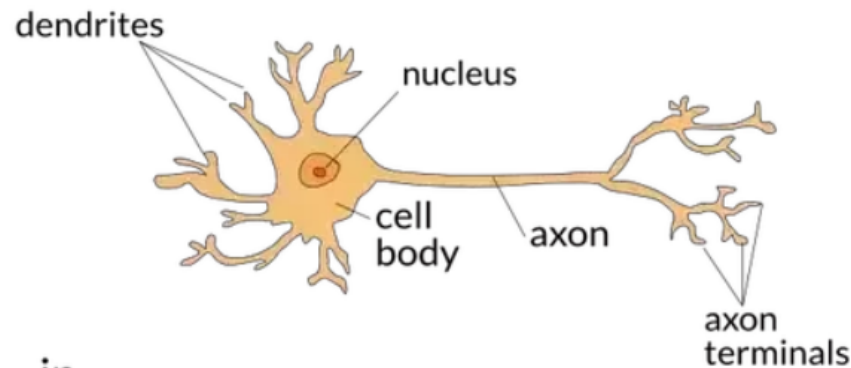
[Image source](#)

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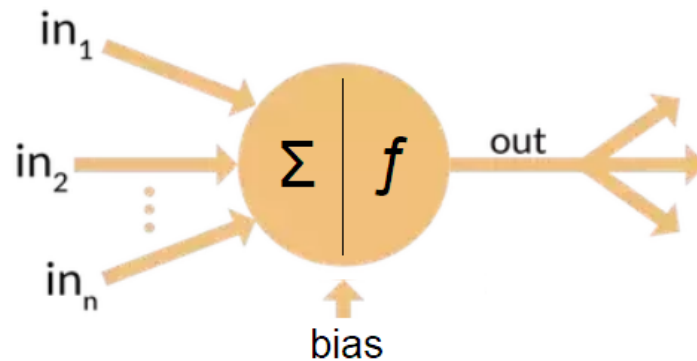
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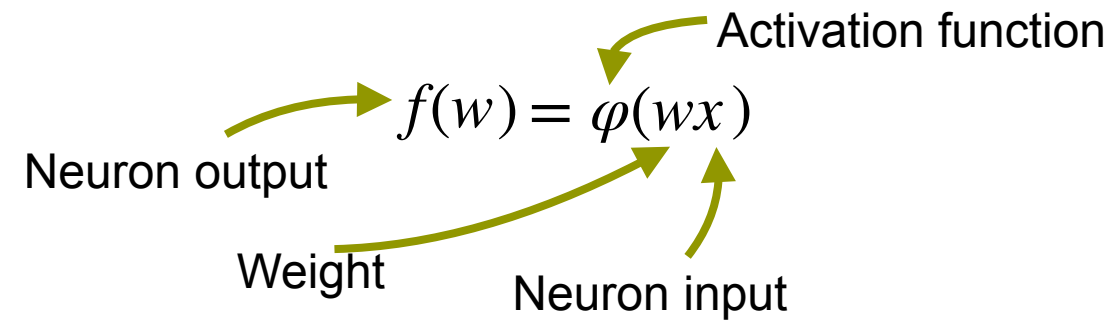


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Simplistic artificial neuron:



e.g. $\varphi(z) = \tanh(z)$

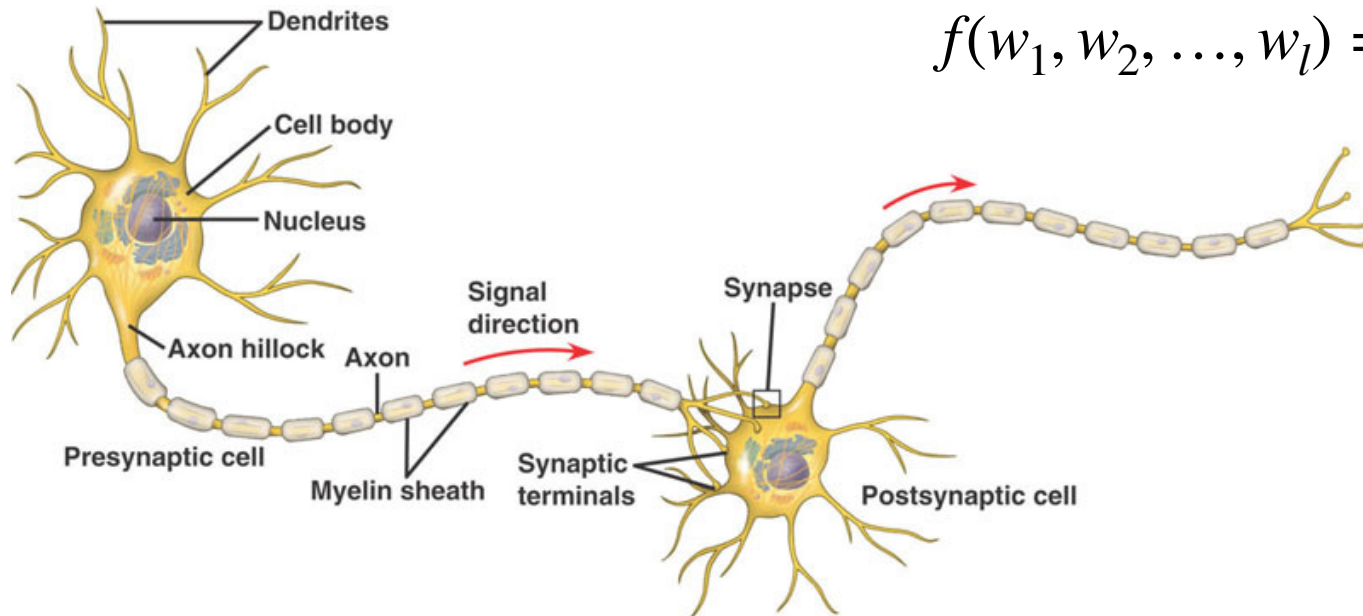
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Introduction

This is maybe a very simple optimisation problem, but optimisation problems are everywhere

Example: training neural networks

In reality we have many artificial neurons that are composed together



$$f(w_1, w_2, \dots, w_l) = \tanh(w_l \cdots \tanh(w_2 \tanh(w_1 x)) \cdots)$$

If we define $h_j(z) = w_j z$, we have

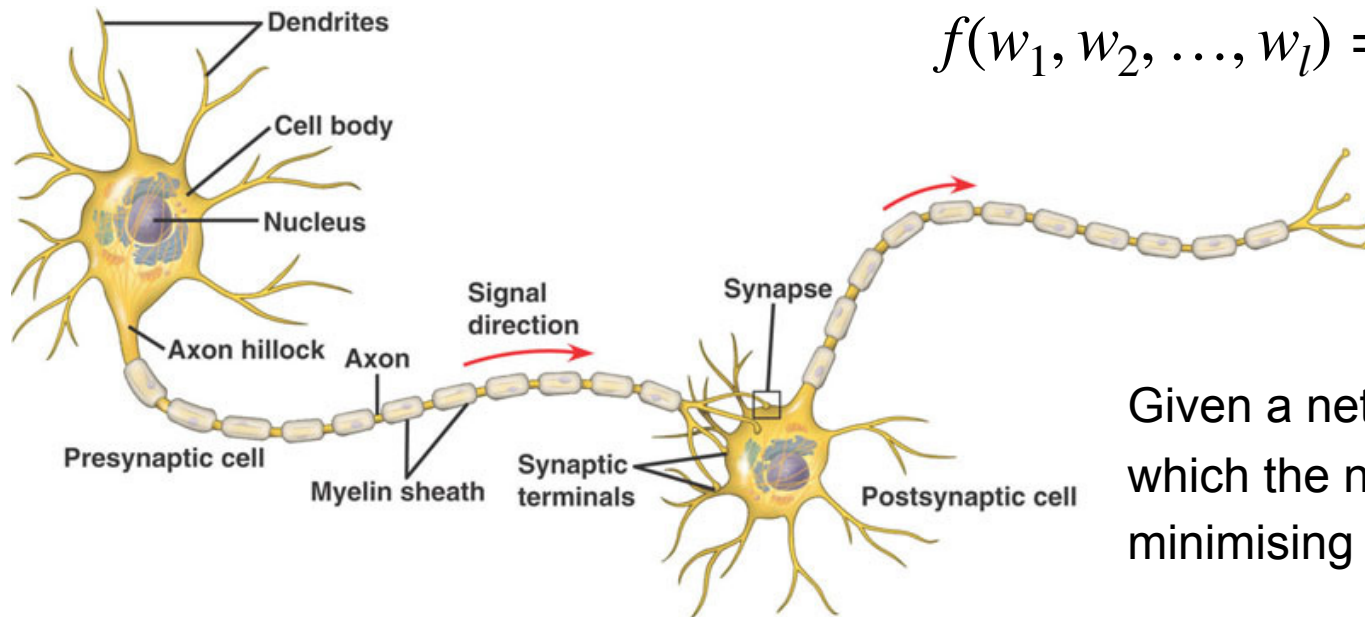
$$f = \varphi \circ h_l \circ \varphi \circ h_{l-1} \circ \dots \circ \varphi \circ h_1$$

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Example: training neural networks

In reality we have many artificial neurons that are composed together



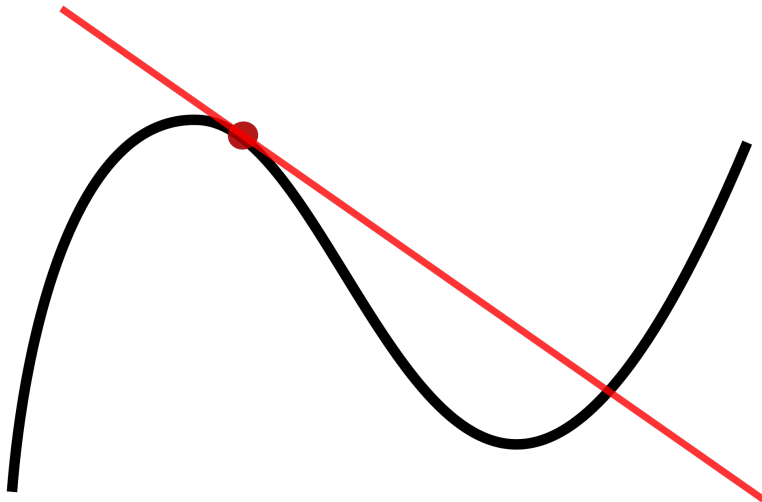
$$f(w_1, w_2, \dots, w_l) = \tanh(w_l \cdots \tanh(w_2 \tanh(w_1 x)) \cdots)$$

Given a network input x , we can find weights $w_1, w_2, \dots, w_l$ for which the neural network output matches a desired output y by minimising an error or loss function between y and f

Introduction

Example: training neural networks

$$g(w_1, \dots, w_l) = \frac{1}{2} \left(\tanh \left(w_l \tanh \left(w_{l-1} \tanh \left(\dots w_2 \tanh(w_1 x) \right) \dots \right) \right) - y \right)^2$$



(usually this is not just done for one pair x and y but many pairs, which are then averaged, but we ignore this today to not make things too complicated)

In order to minimise such a function, optimisation methods usually make use of derivatives of this function, which are computed automatically

Today we learn how to do this with the help of dual numbers

Dual Numbers

What are dual numbers?

Natural numbers

(0,) 1, 2, 3, 4, 5, 6, ...

Integers

..., -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, ...

Rational numbers

Integers and fractions of integers e.g. $1/3$, $-5/11$, 0, $4/2$ (= 2), etc.

Real numbers

Rational numbers and irrational numbers, e.g. $\sqrt{2}$

What are dual numbers?

Real numbers

Rational numbers and irrational numbers, e.g. $\sqrt{2}$

Complex numbers

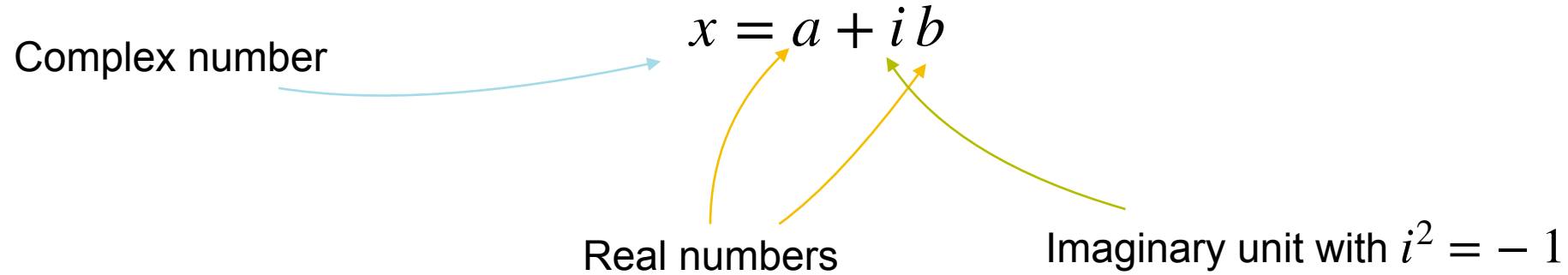
Real numbers and imaginary numbers

Complex number $x = a + ib$

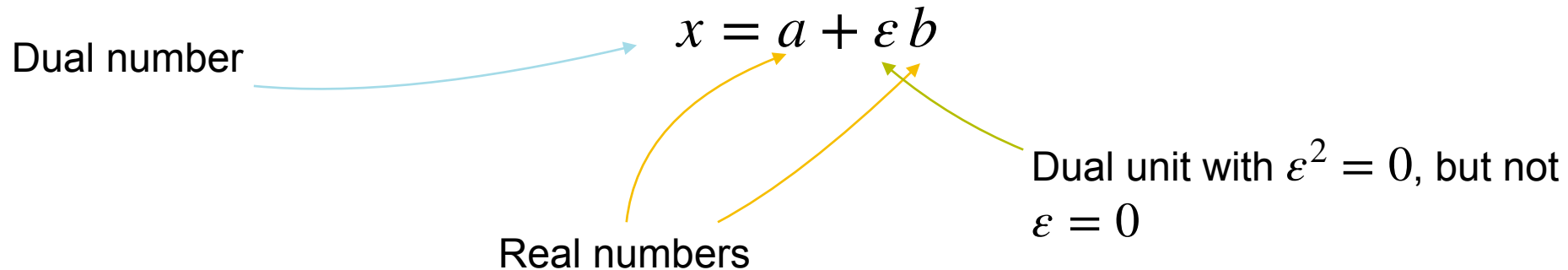
Real numbers Imaginary unit with $i^2 = -1$

The diagram illustrates the structure of a complex number. A light blue arrow points from the text 'Complex number' to the variable x in the equation $x = a + ib$. Two yellow arrows point from the text 'Real numbers' to the variable a and the variable b in the equation. A green arrow points from the text 'Imaginary unit with $i^2 = -1$ ' to the variable i in the equation.

What are dual numbers?

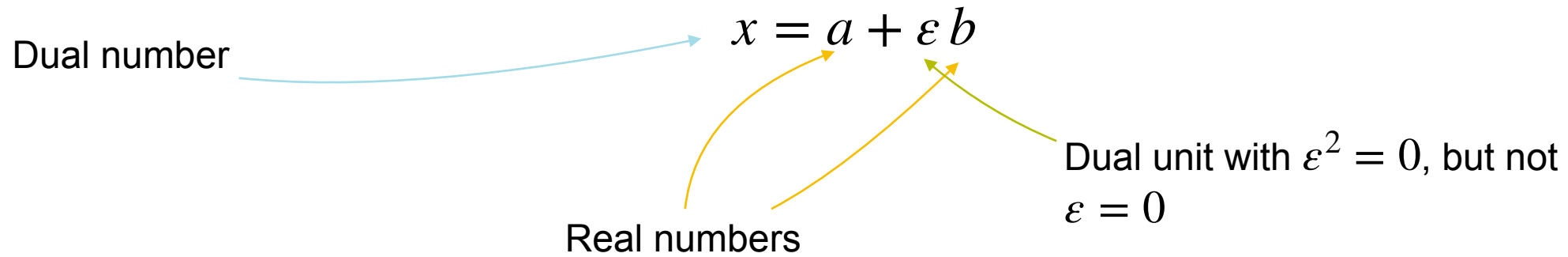


Dual numbers



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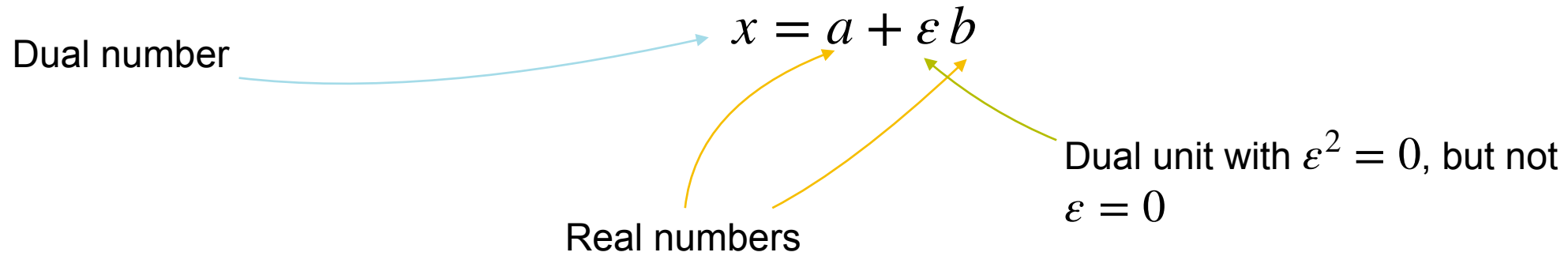


Matrix representation

$$\begin{pmatrix} a & b \\ 0 & a \end{pmatrix} \quad \text{because} \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

What are dual numbers?

Dual numbers



Example

$$xy = (a + \varepsilon b)(c + \varepsilon d) = ac + \varepsilon(ad + bc) + \underbrace{\varepsilon^2 bd}_{=0} = ac + \varepsilon(ad + bc)$$

What are dual numbers good for?

Suppose we look at the polynomial

$$3x^3 - 2x^2 + 1$$

for the dual number $x = a + \varepsilon b$.

We observe

$$\begin{aligned} 3x^3 - 2x^2 + 1 &= 3(a + \varepsilon b)^3 - 2(a + \varepsilon b)^2 + 1 \\ &= 3(a + \varepsilon b)(a^2 + 2ab\varepsilon + \varepsilon^2 b^2) - 2(a^2 + 2ab\varepsilon + \varepsilon^2 b^2) + 1 \\ &= 3(a + \varepsilon b)(a^2 + 2ab\varepsilon) - 2(a^2 + 2ab\varepsilon) + 1 \end{aligned}$$

What are dual numbers good for?

$$\begin{aligned}
 3x^3 - 2x^2 + 1 &= 3(a + \varepsilon b)(a^2 + 2ab\varepsilon) - 2(a^2 + 2ab\varepsilon) + 1 \\
 &= 3a^3 + 6a^2b\varepsilon + 3\varepsilon ba^2 + 6ab^2\varepsilon^2 - 2a^2 - 4ab\varepsilon + 1 \\
 &= 3a^3 - 2a^2 + 1 + (6a^2 + 3a^2 - 4a)\varepsilon b \\
 &= \underbrace{3a^3 - 2a^2 + 1} + \underbrace{(9a^2 - 4a)\varepsilon b}
 \end{aligned}$$

This is the function itself, but in a

What about this function?

What are dual numbers good for?

$$3x^3 - 2x^2 + 1 = \underbrace{3a^3 - 2a^2 + 1} + \underbrace{(9a^2 - 4a)\epsilon b}$$


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What about this function?

The derivative(/slope) of $f(x) = 3x^3 - 2x^2 + 1$ is

$$f'(x) = 9x^2 - 4x$$

Is it a coincidence that $(9a^2 - 4a)b\epsilon$ is the same as $f'(a)b\epsilon$?

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Is it a coincidence that $(9a^2 - 4a)b\epsilon$ is the same as $f'(a)b\epsilon$?

The answer is: No 🙄

You will (hopefully) learn this in Calculus, but there is this concept known as Taylor series, with which we can approximate differentiable functions as follows:

$$f(x) = f(y) + f'(y)(x - y) + \frac{f''(y)}{2}(x - y)^2 + \frac{f'''(y)}{6}(x - y)^3 + \dots$$

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If we choose $x = a + \varepsilon b$ and $y = a$, we observe

$$f(a + \varepsilon b) = f(a) + f'(a)\varepsilon b + \frac{f''(a)}{2}\varepsilon^2 b^2 + \frac{f'''(a)}{6}\varepsilon^3 b^3 + \dots$$

$$= f(a) + \varepsilon b f'(a) \quad \text{because } \varepsilon^2 = 0$$

We can use this to teach a computer to compute derivatives(/slopes) of functions

Live demo!